

CONCRETE NOTES

- 1
- ALL MATERIALS, WORKMANSHIP, HANDLING AND PLACEMENT SHALL BE IN ACCORDANCE WITH THE RELEVANT AUSTRALIAN STANDARDS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
- 2
- BEAM DEPTHS ARE WRITTEN FIRST AND INCLUDE SLAB THICKNESS. BEAMS AND SLABS ARE TO BE POURED TOGETHER UNLESS NOTED OTHERWISE.
- 3
- SIZES OF CONCRETE ELEMENTS DO NOT INCLUDE THICKNESS OF APPLIED FINISHES.
- 4
- HOLES, CHASES OR EMBEDMENT OF PIPES OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL NOT BE MADE IN ANY STRUCTURAL CONCRETE WITHOUT PRIOR APPROVAL OF THE WSP.
- 5
- CONSTRUCTION JOINTS SHALL BE PROPERLY FORMED AND USED ONLY WHERE SHOWN ON THE DRAWINGS OR SPECIFICALLY APPROVED BY THE WSP.
- 6
- ALL EXPOSED CONCRETE CORNERS TO HAVE 15mm CHAMFER U.N.O.
- 7
- CAMBER TO SUSPENDED SLABS SHALL BE POSITIVE UPWARD CAMBER OF 3mm PER 1000mm SPAN UNLESS NOTED OTHERWISE. BEAMS SHALL BE CAMBERED AS SHOWN ON DRAWINGS. NO CAMBER IS REQUIRED TO POST-TENSIONED BEAMS AND SLABS.
- 8
- FORMWORK AND BACKPROPPING SHALL BE DESIGNED, CONSTRUCTED AND STRIPPED IN ACCORDANCE WITH AS 3610. REFER TO ARCHITECTURAL DRAWINGS AND THE SPECIFICATION FOR CLASSES OF SURFACE FINISH.
- 9
- UNLESS OTHERWISE SPECIFIED ALL CONCRETE SHOULD BE 'N' GRADE AND COMPLY WITH THE FOLLOWING PERFORMANCE REQUIREMENTS.
- 9a
- MODULUS OF ELASTICITY 'E' TO COMPLY WITH THE REQUIREMENTS OF AS3600 TABLE 3.1.2.
- 9b
- BASIC CREEP CO-EFFICIENT TO COMPLY WITH THE REQUIREMENTS OF AS3600 TABLE 3.1.2.
- 9c
- IT'SSLAB TO COMPLY WITH THE REQUIREMENTS OF AS3600 C3.1.2.2. NOTING HOWEVER THAT THE CODE DOES NOT PROVIDE PRESCRIPTIVE VALUES FOR EVERY SITE IN AUSTRALIA. FOR THE PURPOSES OF DESIGN WSP HAVE ASSUMED THE FOLLOWING VALUES WHICH ARE TO BE VERIFIED BY THE CONCRETE SUPPLIER PRIOR TO CONSTRUCTION. IF THESE VALUES CANT BE ACHIEVED AS PART OF AN 'N' GRADE MIX, THEN REDESIGN AT COST MAY BE REQUIRED TO ACCOUNT FOR THE MATERIAL AVAILABLE ON SITE. ALTERNATIVELY AN 'S' GRADE MIX CLASSIFICATION MAY BE ADOPTED TO ACHIEVE THE BELOW VALUES. IF THERE ARE ANY ADDITIONAL COSTS ASSOCIATED WITH ADOPTING AN 'S' MIX THEN THESE SHOULD BE RAISED AND APPROVED PRIOR TO CONSTRUCTION COMMENCING.
- 10
- ALL CEMENT IS TO BE 'GP' GENERAL PURPOSE PORTLAND CEMENT OR 'GB' GENERAL PURPOSE BLENDED CEMENT OR TYPE 'SR' SULPHATE-RESISTING CEMENT AS REQUIRED COMPLYING WITH AS3972 UNLESS NOTED OTHERWISE ON THE DRAWINGS. EXTRA RAPID HARDENING SUPERSULPHATED AND HIGH ALUMINA CEMENTS AND CEMENTS CONTAINING CHLORIDE SHALL NOT BE USED. THE USE OF FLY ASH AND/OR SILICA FUME AS A CEMENT SUBSTITUTE, OTHER THAN THAT PROPORTION ALLOWED AS PART OF THE 'GB' CEMENT CONTENT WILL ONLY BE PERMITTED AS PART OF A DESIGNED CONCRETE MIX WHICH HAS BEEN APPROVED IN WRITING BY THE WSP.
- 11
- WHERE NOTED ON PLAN (D) DENOTES SPECIAL DURABLE CONCRETE WHERE THE ELEMENT HAS AT LEAST ONE FACE EXPOSED TO THE WEATHER OR POSSIBLE CORROSIVE ATTACK. (THIS CONCRETE REQUIRES A SPECIAL TOLERANCE FOR THE COVERS OF -0mm + 10mm). SPECIAL PRECAUTIONS ARE REQUIRED TO IMPROVE THE LONG TERM PERFORMANCE OF THESE FACES OF CONCRETE. IN PARTICULAR, NO METAL INSERTS, METAL BAR CHAIRS OR METAL FORM SPACERS OF ANY KIND ARE TO BE PLACED IN THE COVER ZONES WITHOUT THE EXPRESS PERMISSION OF THE WSP. TAKE SPECIAL CARE TO AVOID SCRAP TIE WIRE OR OTHER MATERIAL BEING PRESENT. REFER DRAWINGS FOR LOCATIONS.
- 12
- CONCRETE TESTING SHALL COMPLY WITH THE REQUIREMENTS OF AS1379 FOR PROJECT ASSESSMENT.
- 13
- CONCRETE SLUMP TO BE A MAXIMUM OF 80mm UNLESS NOTED OTHERWISE ON THE DRAWINGS.
- 14
- FREE DROPPING OF CONCRETE FROM A HEIGHT GREATER THAN 1000mm IS NOT PERMITTED.
- 15
- SURFACES RECEIVING GROUT SHALL BE LEFT ROUGH AND FREE OF LAITANCE.
- 16
- CONCRETE MUST BE CURED BY AN APPROVED CURING COMPOUND ACHIEVING A MINIMUM 90% MOISTURE RETENTION IN ACCORDANCE WITH AS3799.
- 17
- REINFORCEMENT IS REPRESENTED DIAGRAMMATICALLY AND NOT NECESSARILY SHOWN IN TRUE PROJECTION OR SCALE.
- 18
- REINFORCEMENT SYMBOLS:
- 18a
- N - NORMAL DUCTILITY CLASS HOT ROLLED DEFORMED BARS OR MESH TO ASINZS 4671 WITH fy=500 MPa.
- 18b
- R - NORMAL DUCTILITY CLASS 250N PLAIN ROUNDED BAR TO ASINZS 4671 WITH fy=250 MPa.
- 18c
- L - LOW DUCTILITY CLASS 500L REINFORCING MESH OR BAR TO ASINZS 4671 WITH fy=500 MPa.
- 18d
- LOW DUCTILITY CLASS L REINFORCEMENT IS NOT TO BE USED OTHER THAN WHERE SHOWN ON DRAWINGS THE NUMBER FOLLOWING THE BAR SYMBOL IS THE NORMAL BAR DIAMETER IN MILLIMETRES.
- 19
- SPLICES IN REINFORCEMENT SHALL BE MADE IN THE POSITIONS SHOWN OR AS OTHERWISE APPROVED BY THE WSP. MINIMUM LAP FOR ALL FABRICS SHALL BE THE SPACING OF TWO TRANSVERSE WIRES PLUS 25mm. GRADE 500N BARS SHALL BE LAPPED IN ACCORDANCE WITH THE STANDARD LAP LENGTH TABLE BELOW IF NOT STATED OTHERWISE ON THE DRAWINGS.
- 20
- HOOKS AND COGS SHALL COMPLY WITH AS 3600 UNLESS OTHERWISE SHOWN ON THE DRAWINGS.
- 21
- CUTTING OR WELDING OF REINFORCEMENT IS NOT ALLOWED WITHOUT THE APPROVAL OF THE WSP.
- 22
- CONDUITS AND OTHER CAST IN ITEMS SHALL BE FABRICATED AND INSTALLED SO THAT NO CUTTING, BENDING OR DISPLACEMENT OF THE REINFORCEMENT FROM ITS PROPER POSITION WILL BE REQUIRED.
- 23
- 3M12-100 TOP DIAGONAL CORNER TRIMMER BARS x 2000mm LONG ARE REQUIRED DIAGONALLY ACROSS ALL RE-ENTRANT CORNERS FOR SLABS ON GROUND. 2M16-100 TOP DIAGONAL CORNER TRIMMER BARS x 1500mm LONG ARE REQUIRED DIAGONALLY ACROSS ALL RE-ENTRANT CORNERS OF OPENINGS IN SUSPENDED SLABS AND WALLS.
- 24
- REINFORCEMENT LENGTHS INDICATED ARE IN MILLIMETRES AND ARE PLAN LENGTH ONLY. TURN DOWNS AND CRANKS ARE NOT INCLUDED IN THE DIMENSION.
- 25
- BAR SHOWN STAGGERED ON PLAN SHALL BE PLACED ALTERNATELY.
- 26
- BAR SHALL BE EVENLY DISTRIBUTED OVER THE WIDTH OF THE STRIP INDICATED ON THE DRAWINGS UNLESS NOTED OTHERWISE.
- 27
- ALL STEEL EMBEDMENTS WITHIN THE CONCRETE COVER ZONE SHALL BE HOT DIP GALVANIZED UNLESS APPROVED OTHERWISE BY THE WSP.
- 28
- CONCRETE SHALL BE SEPARATED FROM SUPPORTING MASONRY BY TWO LAYERS OF MALTHOID (OR AN APPROVED EQUIVALENT). VERTICAL FACES OF CONCRETE SHALL BE KEPT FREE OF ADJOINING SURFACES BY 10mm THICKNESS OF ABLEFLEX (OR AN APPROVED EQUIVALENT) UNLESS NOTED OTHERWISE ON THE DRAWINGS. ALL NON-LOADBEARING WALLS SHALL BE KEPT CLEAR OF THE UNDERSIDE OF SLABS AND BEAMS BY 20mm UNLESS NOTED OTHERWISE ON THE DRAWINGS.
- 29
- THIS TABLE IS TO ONLY BE USED WHERE CONCRETE STRENGTHS AND COVERS ARE NOT NOTED ON STRUCTURAL DRAWINGS.

CONCRETE ELEMENT	CONCRETE STRENGTH (MPa)	BOTTOM COVER (mm)	TOP COVER (mm)	SIDE COVER (mm)
BORED PIERS / PILES	50	-	-	75
PILE CAPS	50	75	75	50
PAD FOOTINGS	32	50	50	50
STRIP FOOTINGS / GROUND BEAMS	32	50	50	50
SLAB ON GROUND	32	40	30	-
SUSPENDED SLABS - INTERIOR	32	25	25	-
SUSPENDED SLABS - EXTERIOR	32	40	30	-
SUSPENDED BEAMS	32	35	35	50
WALLS	32	-	-	40
PRECAST PANELS	REFER TO PRECAST SCHEDULE	-	-	35
COLUMNS	REFER TO COLUMN SCHEDULE	-	-	40

- 30a
- COVER IS THE CLEAR DISTANCE BETWEEN ANY REINFORCING (INCLUDING FITMENTS) AND THE FACE OF THE STRUCTURAL ELEMENT.
- 30b
- FOR ALL EXTERNAL SURFACES, PROVIDE FULLY PLASTIC BAR CHAIRS. TIE WIRE SHALL NOT BE NAILED TO THE FORMS, REINFORCING BARS SHALL NOT BE USED TO KEEP FORMS APART AND A THROUGH TIE SYSTEM SHALL BE USED TO TIE FORMS.
- 30c
- THE COVERS SHALL BE MAINTAINED USING APPROVED BAR CHAIRS. IN SLABS THE BAR CHAIRS SHALL BE AT 800 x 800mm MAXIMUM CENTRES. BAR CHAIRS SHALL BE PROVIDED ALONG THE EDGES OF ALL CONSTRUCTION JOINTS. STOP ENDS SHALL NOT BE USED TO MAINTAIN THE COVERS.
- 30d
- EXTERNAL ELEMENTS ARE THOSE EXPOSED TO WEATHER, RAIN AND WATER PENETRATION AND ARE CLASSIFIED B1 UNLESS NOTED OTHERWISE
- 31
- WELDING OF N500 GRADE REINFORCEMENT TO CAST IN PLATES SHALL COMPLY WITH AS 1554.3. ELECTRODE GRADE E50 SHALL BE USED AND THE TEMPERATURE OF THE WELDED BARS SHALL NOT EXCEED 200 DEGREES CELSIUS. THE BARS SHALL BE BENT TO THEIR INTENDED SHAPE PRIOR TO WELDING.

REINFORCEMENT LAPS

- 1
- THE LAPS PROVIDED BELOW ARE BASED ON 'N500' GRADE REINFORCEMENT
- 2
- LAPS DETAILED ON PLANS / ELEVATIONS TAKE PRIORITY OVER THE LAPS DETAILED BELOW

LAPPED SPLICE - L

U.N.O THE LAPS PROVIDED BELOW APPLY TO;

- TOP REINFORCEMENT ON ELEMENTS > 300mm DEEP
- WALL HORIZONTAL REINFORCEMENT
- WALL HEADER BEAM REINFORCEMENT

db	f _c			
	32	40	50	>= 65
10	500	450	400	350
12	650	550	500	450
16	950	850	750	650
20	1250	1150	1000	900
24	1600	1450	1300	1150
28	1950	1750	1550	1400
32	2300	2100	1850	1650
36	2700	2450	2200	1900
40	3150	2800	2500	2200

LAPPED SPLICE - L2

U.N.O THE LAPS PROVIDED BELOW APPLY TO;

- BOTTOM REINFORCEMENT
- TOP REINFORCEMENT ON ELEMENTS <= 300mm DEEP
- WALL VERTICAL REINFORCEMENT
- STAIR REINFORCEMENT

db	f _c			
	32	40	50	>= 65
10	400	350	300	300
12	500	450	400	350
16	700	650	600	500
20	950	850	800	700
24	1250	1100	1000	900
28	1500	1350	1200	1050
32	1800	1600	1450	1250
36	2100	1900	1700	1500
40	2450	2150	1950	1700

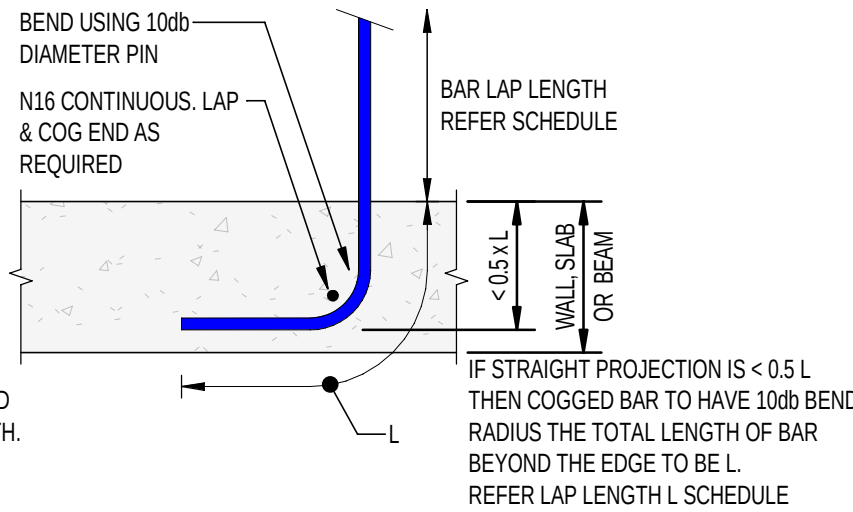
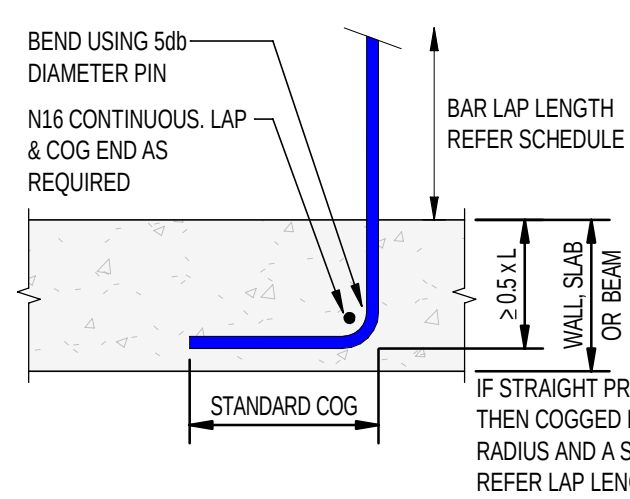
LAPPED SPLICE - L3

U.N.O THE LAPS PROVIDED BELOW APPLY TO;

- COLUMN REINFORCEMENT

db	f _c			
	32	40	50	>= 65
10	350	350	350	350
12	400	400	400	400
16	550	550	550	550
20	650	650	650	650
24	800	800	800	800
28	900	900	900	900
32	1050	1050	1050	1050
36	1200	1200	1200	1200
40	1300	1300	1300	1300

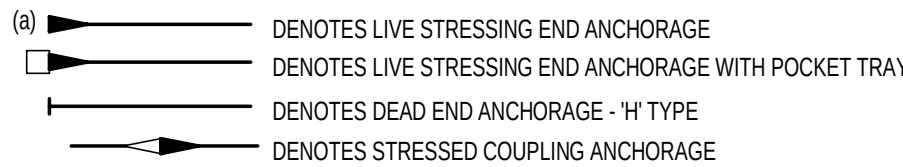
FULL DEVELOPMENT OF COGGED BARS



INFORMATION REFLECTED ON THIS DRAWING REPRESENTS STRUCTURAL DESIGN DEVELOPED FOR D&C TENDER ADVICE ONLY, AND SHOULD NOT BE REGARDED AS COMPLETE. THE D&C CONTRACTOR SHOULD REFER TO ALL ASSOCIATED ARCHITECTURAL AND OTHER DISCIPLINE DRAWINGS, PROJECT BRIEFS AND CLIENT REQUIREMENTS AND MAKE ADEQUATE ALLOWANCES FOR ALL NECESSARY STRUCTURAL WORKS TO MEET THESE REQUIREMENTS, EQUIPMENT SCHEDULES AND COMPLIANCE WITH THE NCC & APPROPRIATE AUSTRALIAN STANDARDS

POST TENSIONING CONCRETE NOTES

- 1
- ALL WORKMANSHIP AND MATERIALS SHALL COMPLY WITH THE REQUIREMENTS OF AS 3600 EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS
- 2
- ALL POST TENSIONED BEAM, SLAB AND BAND BEAM ELEMENTS HAVE BEEN DESIGNED AND DETAILED BASED ON THE VSL SYSTEM. THIS DOES NOT EXCLUDE THE USE OF OTHER EQUIVALENT PRE-STRESSING SYSTEMS PROVIDED FULL ALTERNATE DETAILS ARE SUBMITTED FOR APPROVAL TO THE WSP.
- 3
- THE STRAND USED IN ALL BEAM, SLAB AND BAND BEAM TENDONS SHALL BE 7 WIRE SUPER GRADE STRESS RELIEVED LOW RELAXATION 12.7mm OR 15.2mm DIAMETER STRAND OF 184 / 250 kN MINIMUM BREAKING LOAD TO ASINZS4672.
- 4
- THE POST TENSIONING DESIGN ASSUMES A 70mm OR 90mm WIDE X 19mm DEEP SHEATH OF ZINC COATED STEEL HAVING A FRICTION CO-EFFICIENT OF 0.20 AND A PARASTIC FRICTION CO-EFFICIENT (WOBBLE FACTOR) OF 0.025. SHEATHS HAVING THESE PROPERTIES ARE TO BE USED BY THE CONTRACTOR.
- 5
- THE CONTRACTOR SHALL ENSURE:
- 5a
- TEST CERTIFICATES HAVE BEEN PROVIDED FOR EACH ANCHOR AND/OR STRESSING ATTACHMENT USED ON THE PROJECT.
- 5b
- STRESSING JACKS HAVE BEEN CALIBRATED, THE CALIBRATION IS CURRENT AND CALIBRATION DATA IS AVAILABLE FOR REVIEW.
- 5c
- SHOP DRAWINGS OF THE TENDON LAYOUT, SHOWING PROFILES, ANCHOR DETAILING, NON-PRE-STRESSED REINFORCEMENT AND PROPOSED STRESSING SEQUENCES HAVE BEEN PREPARED AND CERTIFIED BY THE POST-TENSIONING SUB-CONTRACTOR'S ENGINEER.
- 5d
- STRESSING RESULTS ARE RECORDED, CHECKED AND COLLATED. TEST CERTIFICATES, CALIBRATION DETAILS, SHOP DRAWINGS AND STRESSING RESULTS SHALL BE MADE AVAILABLE FOR REVIEW ON REQUEST BY THE WSP.
- 6
- TENDON DESIGNATIONS:



PROFILES

- 7a
- TENDON PROFILES ARE TO BE FROM SOFFIT OF SLAB AND SLAB BAND TO UNDERSIDE OF DUCT BETWEEN ANCHORAGES AND TO THE CENTRELINE AT THE ANCHORAGE
- 7b
- TENDON PROFILES SHALL BE PARABOLIC BETWEEN HIGH AND LOW POINTS U.N.O.
- 7c
- SLAB TENDON PROFILES SHALL BE STRAIGHT THROUGH THE WIDTH OF THE BEAMS.
- 7d
- TRANSVERSE CRACK CONTROL TENDONS SHALL BE LOCATED AS NEAR AS POSSIBLE TO THE MID DEPTH OF THE SLAB AND SHALL BE ANCHORED AT MID DEPTH U.N.O.

STRESSING FORCE

- 8a
- FOR 12.7mm DIA. STRANDS MAXIMUM JACKING FORCE = 156kN/STRAND (85% OF MBL)
- 8b
- FOR 15.2mm DIA. STRANDS MAXIMUM JACKING FORCE = 212kN/STRAND (85% OF MBL)
- 8c
- THE JACKING FORCE SHALL BE THE FORCE IN THE TENDON IMMEDIATELY BEHIND THE ANCHORAGE BEFORE LOCKING OFF.

STRESSING SEQUENCE

- 9a
- JACK TO 25% OF TOTAL JACKING FORCE AFTER CONCRETE HAS ATTAINED A MINIMUM COMPRESSIVE STRENGTH OF 7MPa AND NOT MORE THAN 24 HOURS AFTER CASTING.
- 9b
- JACK TO 100% OF A TOTAL JACKING FORCE NOT LESS THAN 4 DAYS AFTER CASTING AND AFTER CONCRETE HAS ATTAINED A MINIMUM COMPRESSIVE STRENGTH OF 22MPa FOR MONOSTRAND, 25MPa FOR MULTISTRAND.
- 9c
- JACK TO 100% OF A TOTAL JACKING FORCE NOT LESS THAN 4 DAYS AFTER CASTING AND AFTER CONCRETE HAS ATTAINED A MINIMUM COMPRESSIVE STRENGTH OF 22MPa FOR MONOSTRAND, 25MPa FOR MULTISTRAND.
- 10
- THE TENDONS ARE TO BE PRESSURE GROUTED FOR THEIR FULL LENGTH IN ACCORDANCE WITH AS 3600 WITHIN 4 WEEKS OF THE CONCRETE POUR AND AFTER APPROVAL OF STRAND EXTENSIONS BY THE WSP.
- 11
- ALL CONCRETE IN POST-TENSIONED MEMBERS SHALL BE
- 11a
- F_{cp} = 22MPa AT TRANSFER FOR MONOSTRAND
- 11b
- F_{cp} = 25MPa AT TRANSFER FOR MULTISTRAND
- 12
- CONCRETE IS TO HAVE A MAXIMUM SHRINKAGE STRAIN OF 600 MICROSTRAIN TO PREVENT DAMAGE TO STRESSING TENDONS AND REINFORCEMENT.
- 12a
- PROVIDE PLANKS ON BLOCKS ACROSS SLAB BANDS FOR ACCESS DURING CONSTRUCTION
- 12b
- CONCRETE PUMP LINE SHALL BE ADEQUATELY RESTRAINED AND PLACED ON STOOLS OR TIMBER BLOCKS
- 13
- TOP REINFORCEMENT SHALL NOT BE INSTALLED UNTIL AFTER TENDONS HAVE BEEN PLACED.

PRECAST CONCRETE NOTES

- 1
- ALL WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH AS3600 EXCEPT AS VARIED BY THE CONTRACT DOCUMENTS.
- 2
- PRECAST CONCRETE IS TO BE IN ACCORDANCE WITH THE NATIONAL CODE OF PRACTICE FOR PRECAST TILT-UP AND CONCRETE ELEMENTS IN BUILDING COMPONENTS, EXCEPT AS VARIED BY THE CONTRACT DOCUMENTS.
- 3
- FOR LOCATIONS AND NON-STRUCTURAL DIMENSIONS OF PRECAST PANELS REFER TO ARCHITECTURAL DRAWINGS
- 4
- PANELS ARE TO CONSIST OF OFF FORM CONCRETE POURED IN APPROVED MOULDS AS SPECIFIED. FINISH ARE TO BE AS SPECIFIED BY THE ARCHITECT
- 5
- THE CONCRETE IN THE PANELS IS TO HAVE A MINIMUM CHARACTERISTIC STRENGTH f_c OF 40MPa OR AS SPECIFIED ON THE DRAWINGS. THE CONCRETE STRENGTH AT REMOVAL FROM MOULDS IS TO BE A MINIMUM OF 20MPa. ALL PANELS ARE TO BE CONSTRUCTED FROM NORMAL WEIGHT CONCRETE UNLESS NOTED OTHERWISE ON THE DRAWINGS
- 6
- CORBELS TO THE PRECAST PANELS TO BE POURED MONOLITHICALLY WITH THE PANEL
- 7
- WATER CEMENT RATIO = NOT TO EXCEED 0.45 – SLUMP = NOT TO EXCEED 65mm
- 8
- CALCIUM CHLORIDE WILL UNDER NO CIRCUMSTANCES BE PERMITTED. ANY OTHER ADMIXTURES WILL ONLY BE PERMITTED WITH THE WRITTEN CONSENT OF THE WSP
- 9
- ACID ETCHING WILL UNDER NO CIRCUMSTANCES BE PERMITTED. ACID CLEANING WITH DILUTE HYDROCHLORIC ACID OF MAXIMUM CONCENTRATION OF 20 TO 1 MAY BE PERMITTED UNDER CONTROLLED CONDITIONS IF THE CHLORIDE ION PENETRATION AND CONCENTRATION IS WITHIN ACCEPTABLE LIMITS. HIGH PRESSURE WATER JETS ARE TO BE USED TO REMOVE ALL TRACES OF ACID FROM THE SURFACE. FULL DETAILS OF PROPOSED CLEANING PROCEDURE IS TO BE SUBMITTED TO THE WSP FOR WRITTEN APPROVAL PRIOR TO CLEANING
- 10
- THE MAXIMUM TOTAL CHLORIDE ION CONTENT BY MASS OF CEMENT IS NOT TO EXCEED 0.1%. CHEMICAL TEST RESULTS ON TRIAL PANELS TREATED IN THE INTENDED MANNER SHALL BE SUBMITTED TO THE WSP AND APPROVAL GAINED PRIOR TO MANUFACTURE
- 11
- VENEERED CONSTRUCTION AND/OR USE OF DIFFERENT CONCRETE MIX AND/OR USE OF DIFFERENT CONCRETE SUPPLIERS IS NOT PERMITTED
- 12
- DETAILS OF ALL SURFACE TREATMENTS ARE TO BE SUBMITTED TO THE WSP FOR WRITTEN APPROVAL PRIOR TO MANUFACTURE
- 13
- CONCRETE COVER SHALL BE AS NOTED ON THE DRAWINGS. UNLESS OTHERWISE SPECIFIED ADOPT COVER TO THE EXPOSED FACE OF 35mm AND THE INTERNAL FACE OF 25mm, AS A MINIMUM. TOLERANCE + 5mm, - 0mm UNLESS NOTED OTHERWISE
- 14
- LIFTING FERRULES ARE TO BE DESIGNED AND DETAILED BY THE PRECAST MANUFACTURER TO SUIT THE METHOD OF PRECAST MANUFACTURE AND ERECTION
- 15
- TYPICAL LIFTING FERRULES ARE TO BE AT LEAST 24mm DIA. HOT DIPPED GALVANIZED STEEL FERRULES FITTED WITH A 'J' BAR ANCHOR CAPABLE OF DEVELOPING AT LEAST THE FULL DESIGN CAPACITY OF THE BOLT. LIFTING FERRULES ARE NOT SHOWN ON DESIGN DRAWINGS
- 16
- LIFTING FIXINGS SHALL NOT BE LOCATED IN THE FACE WHICH IS EXPOSED TO VIEW IN THE FINAL CONDITION AND AFTER USE SHALL BE PROTECTED TO AVOID CORROSION AND STAINING
- 17
- FERRULES THAT WOULD BE EXPOSED ON COMPLETION OF ERECTION ARE TO BE SET 25mm BELOW FINISHED LEVEL AND THE RECESS TO BE GROUTED ON COMPLETION
- 18
- ALL CONNECTION MEMBERS AND FIXINGS ARE TO BE HOT DIPPED GALVANIZED
- 19
- CLOSED CELL POLYETHYLENE FOAM ROD COMPRESSION SEALS (25% MIN. COMPRESSION) OR APPROVED EQUIVALENT TO BE USED AS SHOWN ON DETAILS TO CONTAIN INSITU GROUT AND TO EXCLUDE CONCRETE SCREED IN JOINTS BETWEEN PANEL AND INSITU FLOOR
- 20
- FOR DETAILS OF FIRE PROTECTION AND WEATHERPROOFING OF JOINTS REFER TO ARCHITECTS DRAWINGS
- 21
- PRECAST MANUFACTURER TO PROVIDE A SET OF FULLY DETAILED SHOP DRAWINGS INCLUDING DETAILS OF ALL INSERTS, FIXINGS, ANCHORS, BOLTS, FERRULES, PENETRATIONS, LIFTING DEVICES, ETC. TO WSP FOR APPROVAL
- 22
- THE CONTRACTOR IS TO PROVIDE TEMPORARY SUPPORTS DURING THE ERECTION OF THE PRECAST PANELS AS REQUIRED TO ENSURE STABILITY OF THE PRECAST UNITS UNTIL ALL THE PERMANENT FIXINGS BOTH TOP AND BOTTOM HAVE BEEN INSTALLED
- 23
- GROUT TO BE USED SHALL BE NON-SHRINK AND HAVE A MINIMUM 28 CHARACTERISTIC STRENGTH OF THE PRECAST PANEL STRENGTH UNDER BUT NOT LESS THEN 40MPa
- 24
- THE PRECAST PANELS HAVE BEEN DESIGNED FOR IN SERVICE CONDITIONS ONLY (IE. LOADS THE PRECAST PANELS ARE SUBJECT TO AFTER ERECTION ON SITE)
- 25
- THE PRECAST MANUFACTURER IS TO PROVIDE COMPUTATIONS TO THE WSP FOR REVIEW OF STRESSES INDUCED DURING REMOVAL FROM THE MOULD, HANDLING, LIFTING, TRANSPORTATION AND ERECTION. ADDITIONAL REINFORCEMENT REQUIRED FOR THESE CONDITIONS IS THE RESPONSIBILITY OF THE PRECAST MANUFACTURER AND SHALL BE ALLOWED FOR IN THE TENDER
- 26
- PANELS CALLED UP AS SIMILAR ARE ONLY STRUCTURALLY SIMILAR. ARCHITECTURAL DIFFERENCES SUCH AS DIMENSIONS, GROOVES, REBATES, FERRULES, MINOR PENETRATIONS, ETC. MAY EXIST BETWEEN PANELS WHICH ARE STRUCTURALLY SIMILAR. REFER ARCHITECTS DRAWINGS FOR DETAILS. DIMENSIONS ON THE STRUCTURAL DRAWINGS ARE THE MINIMUM STRUCTURAL DIMENSIONS. ANY ARCHITECTURAL FINISHES SUCH AS GROOVES ARE IN ADDITION TO THIS THICKNESS.
- 27
- GROUT DONUTS SHALL NOT BE INSTALLED AROUND DOWELS UNLESS EXPLICITLY APPROVED BY WSP.

FORMWORK NOTES

- 1
- FORMWORK SHALL BE DESIGNED AND CONSTRUCTED IN ACCORDANCE WITH AS 3610 EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
- 2
- PRIOR TO PLACEMENT OF CONCRETE, THE CONTRACTOR SHALL HAVE THE FORMWORK INSPECTED BY THE FORMWORK DESIGN ENGINEER AND SUPPLY THE SUPERINTENDENT WITH A WRITTEN CERTIFICATION THAT THE FORMWORK COMPLIES WITH THE REQUIREMENTS OF AS 3610.
- 3
- THE CLASS AND COLOUR OF THE CONCRETE SURFACE FINISH SHALL BE AS SPECIFIED ON THE ARCHITECTURAL DRAWINGS AND/OR THE SPECIFICATION. APPROVAL OF THE CONCRETE MIX DOES NOT ABSOLVE THE CONTRACTOR OF THE NEED TO COMPLY WITH THE REQUIRED CONCRETE COLOUR REQUIREMENTS.
- 4
- EITHER BACKPROPPING OR AN UNDISTURBED SUPPORT SYSTEM OF FORMWORK IS TO BE ADOPTED. A RE-SHORING SYSTEM IS NOT TO BE USED.
- 5
- THE FOLLOWING REQUIREMENTS SHALL BE INCORPORATED INTO THE FORMWORK DESIGN AND/OR ALLOWED FOR BY THE FORMWORK SUB-CONTRACTOR AS APPROPRIATE:
- 5a
- MINIMUM FORMWORK STRIPPING TIMES ARE TO BE AS FOLLOWS:
- 5b
- VERTICAL SURFACES MAY BE STRIPPED OF FORMWORK WHEN THE MINIMUM MEAN COMPRESSIVE STRENGTH OF CONCRETE f_{cm} HAS REACHED 5.0 MPa. IF THE STRENGTH CANNOT BE ACCURATELY ASSESSED VIA THE USE OF TEST CYLINDERS STORED ON SITE AND CURED IN A SIMILAR MANNER TO THE PERMANENT STRUCTURE THEN THE MINIMUM FORMWORK STRIPPING TIME IS TO BE 2 DAYS PROVIDED THE AVERAGE AMBIENT TEMPERATURE OVER THAT PERIOD IS BETWEEN 12 AND 20 DEGREES CELSIUS.
- 5c
- SOFFITS OF BEAMS AND SLABS MAY BE STRIPPED OF FORMWORK AND BACKPROPPED WHEN THE MINIMUM MEAN COMPRESSIVE STRENGTH OF CONCRETE f_{cm} HAS REACHED 22MPa. IF THE STRENGTH CANNOT BE ACCURATELY ASSESSED VIA THE USE OF TEST CYLINDERS STORED ON SITE AND CURED IN A SIMILAR MANNER TO THE PERMANENT STRUCTURE THEN THE MINIMUM FORMWORK STRIPPING TIME IS TO BE 6 DAYS PROVIDED THE AVERAGE AMBIENT TEMPERATURE OVER THAT PERIOD IS BETWEEN 12 AND 20 DEGREES CELSIUS.
- 5d
- REMOVAL OF FORMWORK SUPPORT (PROPS) TO BEAM AND SLAB SOFFITS MAY BE UNDERTAKEN WHEN THE MINIMUM MEAN COMPRESSIVE STRENGTH OF THE CONCRETE f_{cm} HAS REACHED 28MPa AND A MINIMUM 14 DAYS. IF THE STRENGTH CANNOT BE ACCURATELY ASSESSED VIA THE USE OF TEST CYLINDERS STORED ON SITE AND CURED IN A SIMILAR MANNER TO THE PERMANENT STRUCTURE THEN THE MINIMUM FORMWORK STRIPPING TIME IS TO BE 18 DAYS PROVIDED THE AVERAGE AMBIENT TEMPERATURE OVER THAT PERIOD IS BETWEEN 12 AND 20 DEGREES CELSIUS.
- 6
- THE FORMWORK DESIGN SHALL ALLOW FOR 4 kPa OF STACKED MATERIALS ON ANY AREA OF THE FLOOR SYSTEM. ALL OTHER FORMWORK DESIGN LOADS SHALL BE SPECIFIED IN SECTION 4.4 OF AS 3610.
- 7
- THE FLOOR SYSTEM IS TO ATTAIN A MINIMUM MEAN COMPRESSIVE STRENGTH OF CONCRETE f_{cm} OF 15MPa PRIOR TO STACKING OF MATERIALS ON THAT LEVEL OF THE PERMANENT STRUCTURE.
- 8
- A MINIMUM OF THREE 3 LEVELS OF FORMWORK &/OR BACKPROP SUPPORTS ARE REQUIRED. BASED UPON A 6 DAY CYCLE TIME AND PROVIDED THE AVERAGE AMBIENT TEMPERATURE OVER THAT PERIOD IS BETWEEN 12 AND 20 DEGREES CELSIUS.
- 9
- HORIZONTAL LOADS ON FORMWORK ARE TO BE RESISTED BY THE FORMWORK BRACING. STRUTTING OF THE FORMWORK FROM THE PERMANENT STRUCTURE TO RESIST HORIZONTAL LOADS IS NOT PERMITTED.

FORMWORK NOTES

- 10
- POST-TENSIONING OF THE BEAMS MAY CAUSE CONTRACTION OF THE CONCRETE HENCE THE FORMWORK SHALL BE DESIGNED WITH ADEQUATE PROVISION TO ALLOW FOR ADJUSTMENT AND REMOVAL OF THE FORMS. POST-TENSIONING MAY ALSO CAUSE LIFTING OF THE FLOOR SYSTEM HENCE THE FORMWORK SHALL BE DESIGNED TO ALLOW FOR READJUSTMENT OF PROPS.
- 11
- REFER TO THE DRAWINGS FOR LOCATIONS OF CONSTRUCTION JOINTS.
- 12
- THE SEQUENCE OF CONCRETE PLACEMENT FOR THE WORKS IS TO BE SPECIFIED BY THE CONTRACTOR UNLESS NOTED OTHERWISE ON THE DRAWINGS.
- 13
- PROPPING OF COMPOSITE CONSTRUCTION IS TO BE IN ACCORDANCE WITH MANUFACTURER REQUIREMENTS AND AS SPECIFIED ON THE DRAWINGS.
- 14
- DESIGN LOADS FOR THE PERMANENT STRUCTURES ARE NOTED ON THE DRAWINGS.
- 15
- A GEOTECHNICAL REPORT No. 1-29688 BR. 1.1 BY SOIL SURVEYS ENGINEERING IS AVAILABLE FOR GUIDANCE AS TO THE FOUNDATION MATERIAL AVAILABLE FOR THE SUPPORT OF FORMWORK.
- 16
- DETAILS OF THE REQUIREMENTS OF PERMANENT FORMWORK SHALL BE AS SPECIFIED IN THE CONTRACT DOCUMENTS

EXPOSED VISUAL CONCRETE NOTES

- 1
- GENERAL - ELEMENTS OF THE CONCRETE STRUCTURE, AS IDENTIFIED BY THE ARCHITECT, ARE EXPOSED TO VIEW IN THE FINISHED BUILDING. THESE NOTES ARE TO BE READ IN ADDITION TO THE GENERAL CONCRETE, FORMWORK, AND POST-TENSIONING NOTES AND APPLY TO ANY CONCRETE IDENTIFIED AS BEING EXPOSED IN THE FINISHED BUILDING.
- 2
- METHOD STATEMENT – THE CONTRACTOR IS TO SUBMIT A METHOD STATEMENT FOR THE CONCRETE WORKS THAT SHALL ADDRESS EACH OF THE ISSUES RAISED IN THIS SET OF NOTES.
- 3
- SAMPLE – PRIOR TO COMMENCING THE MAIN STRUCTURE, THE CONTRACTOR IS TO PRODUCE SAMPLES COMPRISING A FULL-HEIGHT COLUMN, A WALL PANEL, AND A SECTION OF BEAM THAT IS REPRESENTATIVE OF THE FINAL WORKS AND USE THE METHODOLOGY OUTLINED IN THE CONTRACTOR'S METHOD STATEMENT. THE SAMPLE IS TO USE THE FORMWORK, CONCRETE, REINFORCEMENT, PLACEMENT AND STRIPPING METHODS THAT WILL BE FOLLOWED THROUGHOUT THE WORKS. THE SAMPLE WILL BE INSPECTED, AND IF AGREED TO COMPLY WITH THE ARCHITECT'S REQUIREMENTS MAY BE INCORPORATED INTO THE WORKS. THE SAMPLE IS THEN TO BE PROTECTED SO IT CAN BE REFERENCED LATER IF REQUIRED. THE SAMPLE MAY BE A PART OF THE WORKS.
- 4
- APPEARANCE – THE APPEARANCE OF THE CONCRETE IS TO BE AS DEFINED BY THE ARCHITECT. THE INTENT IS TO ACHIEVE A FINISH BETTER THAN CLASS 2, BUT NOT TO THE STANDARD OF CLASS 1. THESE ELEMENTS ARE INTENDED TO BE EXPOSED 'AS STRUCK' FROM THE FORMWORK. THEY ARE NOT TO BE RUBBED, REPAIRED, RENDERED OR OTHERWISE TREATED EXCEPT AS SPECIFIED BY THE ARCHITECT.
- 5
- COLOUR – THE CONCRETE COLOUR IS TO BE UNIFORM THROUGHOUT ELEMENTS OF EACH TYPE (FLOORS, COLUMNS, AND WALLS).
- 6
- CONCRETE – TO ENSURE CONSISTENCY OF COLOUR AND APPEARANCE, THE CONCRETE AND ITS CONSTITUENTS ARE TO BE OBTAINED FROM ONE SOURCE FOR THE DURATION OF THE PROJECT. THIS INCLUDES ANY CONSTITUENTS THAT IMPACT ON THE COLOUR SUCH AS CEMENT, AND FINE AND COARSE AGGREGATES. THE MIX DESIGNS AGREED AND USED IN THE SAMPLES ARE NOT TO BE VARIED FOR THE DURATION OF THE PROJECT.
- 7
- COMPACTION - THE CONCRETE IS TO BE THOROUGHLY COMPACTED, WITH A SURFACE THAT IS SMOOTH AND UNIFORM. HONEYCOMBING OR GROUT LEAKAGE OF GREATER THAN 25mm WIDTH ON AN EDGE OR 50mm ON THE FACE OF AN ELEMENT WILL NOT BE ACCEPTED. SOME BLOW-HOLES, TO AN EXTENT GREATER THAN THAT DEFINED AS CLASS 1, MAY BE ACCEPTED.
- 8
- FORMWORK DESIGN – THE FORMWORK MAY BE CONSTRUCTED USING MATERIALS AND METHODS APPROPRIATE FOR THE PRODUCTION OF CLASS 2 FINISH. THE FORMWORK IS TO BE CONSTRUCTED WITH JOINTS AND TIE-BOLTS IN LOCATIONS AGREED WITH THE ARCHITECT.
- 9
- FORMWORK REPAIR – IN THE EVENT OF DAMAGE TO AN ELEMENT OF THE FORMWORK OCCURRING DURING THE DURATION OF THE PROJECT, THE ENTIRE SHEET OF PLYWOOD IS TO BE REPLACED. LOCAL REPAIRS ARE NOT TO BE CARRIED OUT TO FORMWORK IN VISUALLY EXPOSED AREAS. SHOULD THE SURFACE OF THE FORMWORK BECOME DEGRADED TO A DEGREE THAT IT CAN NO LONGER PRODUCE THE STANDARD OF FINISH AGREED FOR THE SAMPLE, THEN THE FORMWORK FACING IS TO BE REPLACED.
- 10
- FORMWORK STRIKING – THE FORMWORK IS TO BE DESIGNED AND CONSTRUCTED SUCH THAT IT CAN BE STRUCK IN A MANNER THAT DOES NOT DAMAGE THE SURFACE FINISH, CORNERS, CHAMFERS ETC. THE FORMWORK IS TO BE STRUCK IN A MANNER THAT DOES NOT DAMAGE THE SURFACE FINISH OF THE CONCRETE. THE FORMWORK IS NOT TO BE STRUCK UNTIL THE CONCRETE HAS GAINED SUFFICIENT STRENGTH SO THAT THE SURFACE WILL NOT BE DAMAGED DURING STRIKING.
- 11
- RELEASE AGENT – A NON-STAINING RELEASE AGENT IS TO BE SELECTED AND USED FOR THE CONSTRUCTION OF THE SAMPLE. THIS RELEASE AGENT IS TO BE USED CONSISTENTLY THROUGHOUT THE PROJECT.
- 12
- CURING / SEALING – AS SOON AS IS PRACTICAL AFTER STRIKING OF THE FORMWORK, THE CONCRETE SURFACE IS TO RECEIVE AN APPROVED NON-STAINING SEALER AS AGREED WITH THE ARCHITECT. THIS SEALER IS NOT TO VARY FROM THAT USED IN THE SAMPLE.
- 13
- PROTECTION – EXPOSED VISUAL CONCRETE SURFACES ARE TO BE PROTECTED FROM ACCIDENTAL DAMAGE, PAINT MARKING, TEMPORARY FIXINGS ETC FOR THE DURATION OF THE WORKS.